

Session 4 Date: April 29, 2026 (2-3 hours)

Objective: First demonstrate CC by using label propagation then compare the three listed algorithms to the response files.

Attempts made:

CC was the most different for the three algorithms for the reasons you have listed. My first attempt only scanned in-edges in Process() which led to wrong labels as labels could only spread in one direction. The component IDs were not getting to the correct minimums.

Solutions/Workarounds:

Had to scan in and out edges to make labels spread no matter what direction the edges are pointing:

```
cppfor (uint32_t i = g.in_offsets[v]; i < g.in_offsets[v+1]; i++)
```

```
    best = min(best, prev[g.in_edges[i]]);
```

```
for (uint32_t i = g.offsets[v]; i < g.offsets[v+1]; i++)
```

```
    best = min(best, prev[g.edges[i]]);
```

Every vertex gets to take whatever minimum label is surrounding it. They then spread their new labels and iterate. They stop as soon as there are no new labels. I also made results to write to results files for the three algorithms to see if they match the correct answer files.

Commits:

added the CC algorithm, writing output files for all three algorithms, verified the output files against the expected answers

Experiments/Analysis:

Tested CC on the disconnected graph (0 1 1 / 1 2 1 / 3 4 1):

CC labels:

vertex 0 = 0

vertex 1 = 0

vertex 2 = 0

vertex 3 = 3

vertex 4 = 3

CC serial vs parallel match ✓

It was predictable for vertices 0, 1, and 2 to all fall into the same component and receive the same label of 0, and for vertices 3 and 4 to fall into the same component and receive the label 3. I then executed the same process on the complete LiveJournal dataset, and compared the output to the answer files. Each of the three returned no output, meaning that there were no differences. I then concluded that everything matched exactly.

Learning outcome:

Forcing CC to only scan in-edges meant that labels would get “stuck” and would not spread throughout the given component. The resolution to the “stuck” labels was the implement scan VS post in both directions. The implementation of the CC algorithm on the 4.8 million vertex graph, and the absence of difference against all three algorithms was beneficial in determining the implementation stability.